

The University of Melbourne
School of Computing and Information Systems
COMP90086 Computer Vision, 2024 Semester 2

Assignment 2: Image Recognition with CNNs

Due: 11:59pm, 14 Sept 2025
Submission: Source code (in Jupyter Notebook) and written responses (as .pdf)
Marks: The assignment will be marked out of 7 points, and will contribute 7% of your total mark.

In this assignment, you will build and evaluate a convolutional neural network for an image recognition task. The goal of the task is to recognize a small square patch containing a letter (T or L) embedded in a background of random rectangles. The images are 28 x 28 grayscale images. The letter patch is 5, 7, or 9 pixels square and may appear anywhere in the image. The letter T or L is always brighter than its background patch.

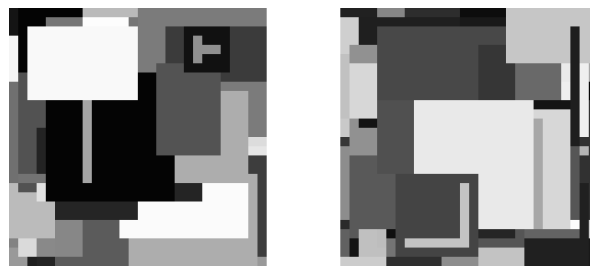


Figure 1: Example images from the provided dataset: T (left) and L (right).

1. The image recognition task [1.5 marks]

Examine the images in the training folder and explain what makes this task challenging from a computer vision perspective. What are the features that the model will need to detect? What types of within-class variation will the classifier need to ignore? If you were to hand-design a convolutional filter or set of filters for this task, what filter(s) might be suitable? Sketch the filter design in your write-up.

2. CNN architecture [2.5 marks]

Implement a small convolutional neural network (CNN) to perform this task. Most of the code has been provided in the Jupyter notebook, but you will need to add the convolutional layers for the network. (You might also want to include a maxpool layer after convolution.) Try to find a network which can perform the task with a reasonably small number of parameters. You should experiment with:

- Number of layers (depth) of the network. For simplicity, stick to 1 or 2 conv layers only, don't try networks with depth greater than 2.
- Number of kernels (width)
- Kernel size

In your write-up, state the architecture of your final model. Show your experiment results and use these to justify your architectural choices. Show the train and test curves for your final model.

You may edit other parts of the provided Notebook code as needed to train your architectures (for example, change the mini-batch size or add dropout or other regularisation). However, you should note these changes in your write-up. Your final model's architecture should not differ substantially from the framework given in the spec: it should have one or two conv layers, followed by one dense layer and a binary classification layer.

3. Kernel visualisation [1 mark]

Visualise the convolutional kernels in your final trained model from question 2. Did the model learn the filters you would expect it to learn, based on your understanding of the task (question 1)? Why or why not?

4. Where is the model looking? [2 marks]

Which image pixels does the model use to make the classification decision? In lecture, we discussed a few methods to investigate how much weight a model puts on each part of an image when making a classification. Implement one of these methods for your final trained model from question 2. Show the results for a few example images of each class and discuss – does the model seem to be looking in the right place when it classifies these images?

Submission

You should make two submissions on the LMS: your code and a short written report explaining your method and results. The response to each question should be no more than 500 words.

Submission will be made via the Canvas LMS. Please submit your code and written report separately under the **Assignment 1** link on Canvas.

- Your **code** submission should include the completed Jupyter Notebook template with your code and any additional files (other than the files we provided) that we would need to run your code. Please include the cell output in your notebook submission if possible.
- Your written **report** should be a .pdf with your answers to each of the questions. The report should address the questions posed in this assignment and include any images, diagrams, or tables required by the question.

Evaluation

Your submission will be marked on the correctness of your code/method, including the quality and efficiency of your code. For efficiency, your image processing steps should use convolution and/or matrix operations (your code should NOT include loops that iterate over all image pixels). You should use built-in Python functions where appropriate and use descriptive variable names. Your written report should clearly address the questions, and include all of the specific outputs required by the question (e.g., images, diagrams, tables, or responses to sub-questions).

Late submission

The submission mechanism will stay open for one week after the submission deadline. Late submissions will be penalised at 10% of the total possible mark per 24-hour period after the original deadline. Submissions will be closed 7 days (168 hours) after the published assignment deadline, and no further submissions will be accepted after this point.

To request an extension on this assignment, please see the [FEIT extension policy](#) and follow the steps below:

- **To request an extension of 1-3 days (24-72 hours)**, complete the online declaration form available at the link above.
- **To request an extension of 4-7 days**, please apply for Special Consideration.

Please note that extension requests should be made before the assignment is due. Late extension requests can only be granted through Special Consideration. The longest possible extension for this assignment is 7 days (168 hours).

Updates to the assignment specifications

If any changes or clarifications are made to the project specification, these will be posted on the LMS.

Academic misconduct

You are welcome — indeed encouraged — to collaborate with your peers in terms of the conceptualisation and framing of the problem. For example, we encourage you to discuss what the assignment specification is asking you to do, or what you would need to implement to be able to respond to a question.

However, sharing materials — for example, showing other students your code or colluding in writing responses to questions — or plagiarising existing code or material (including materials generated by chatGPT or other AI) will be considered cheating. Your submission must be your own original, individual work. We will invoke University's [Academic Misconduct policy](#) where inappropriate levels of plagiarism or collusion are deemed to have taken place.